

Model Performance Metrics

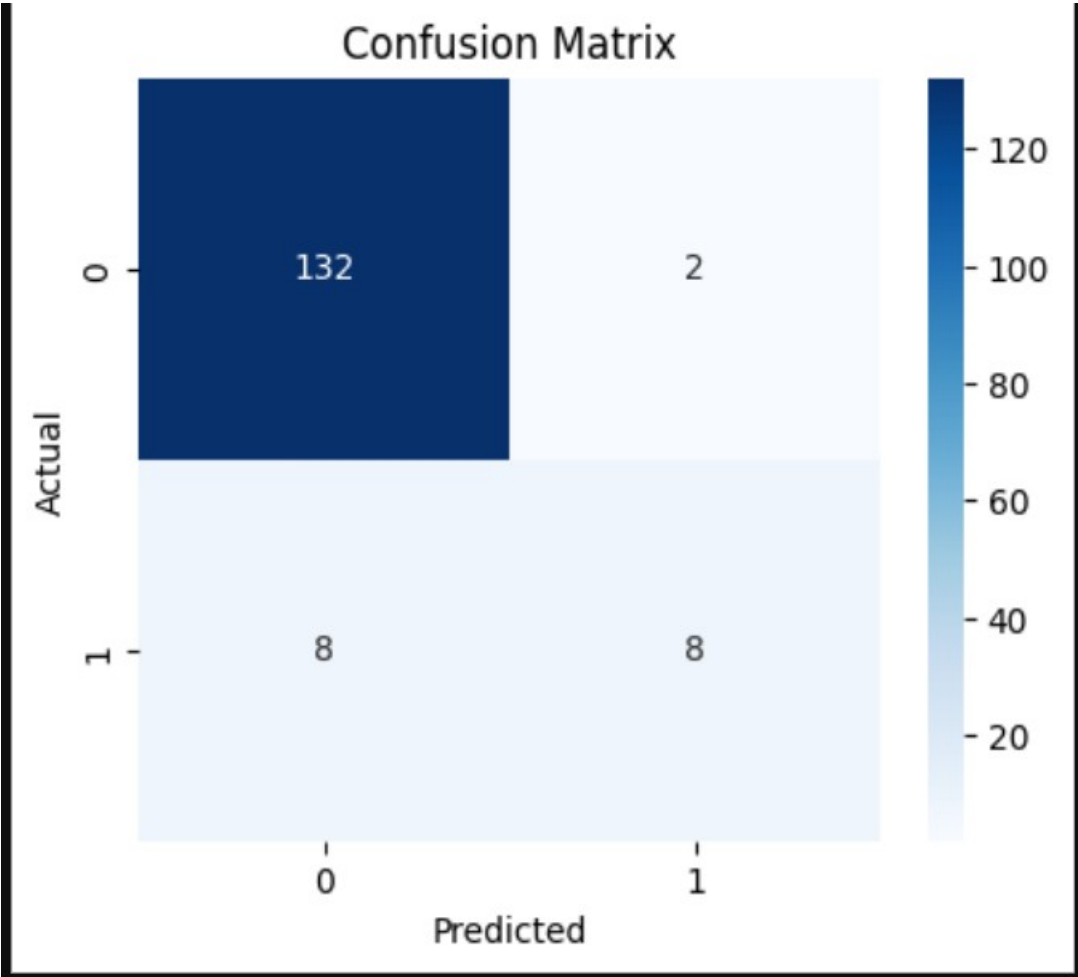
- Accuracy Score: 0.9333
- ROC AUC Score: 0.9837

Classification Report

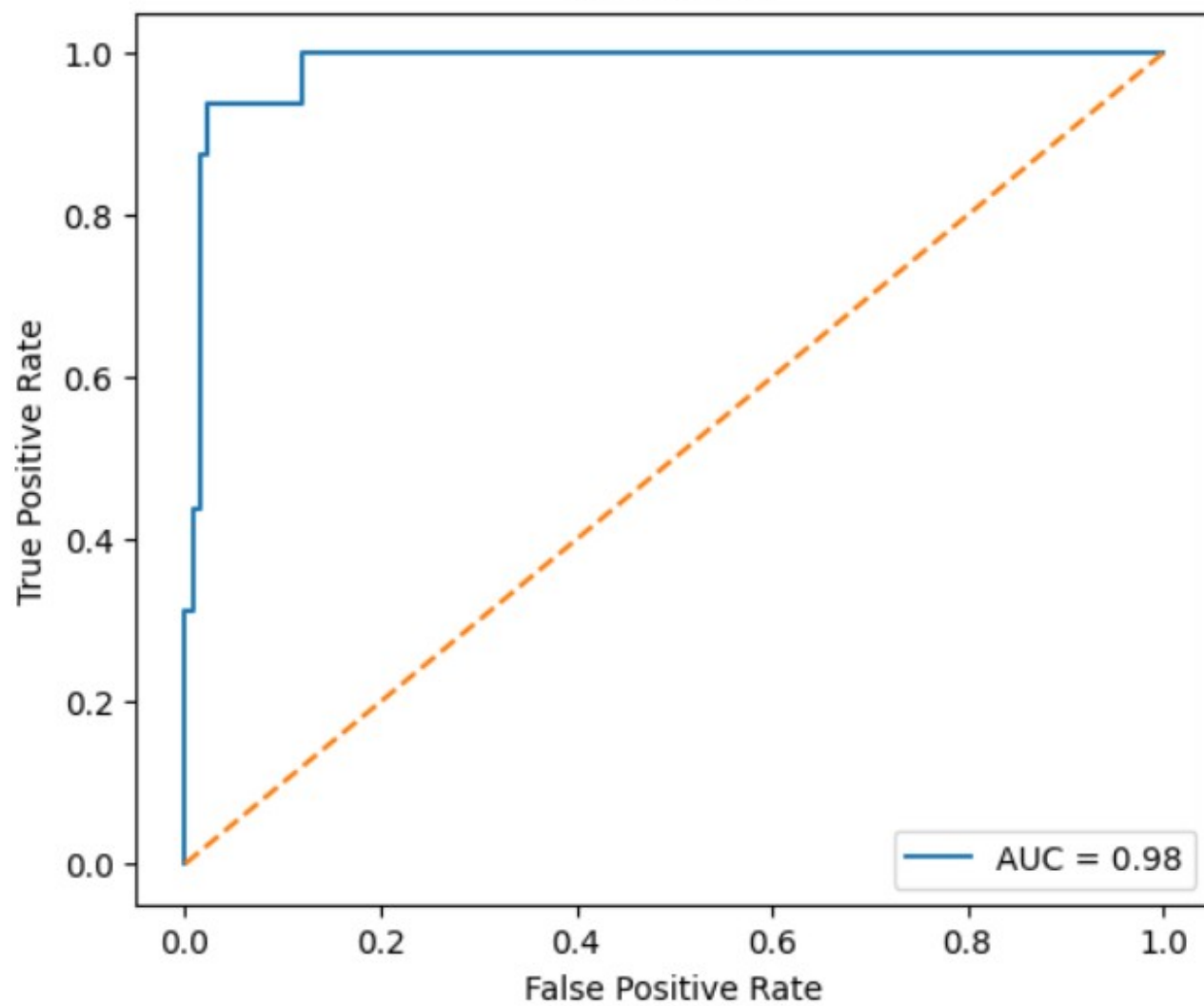
	precision	recall	f1-score	support
0	0.94	0.99	0.96	134
1	0.80	0.50	0.62	16
accuracy			0.93	150
macro avg	0.87	0.74	0.79	150
weighted avg	0.93	0.93	0.93	150

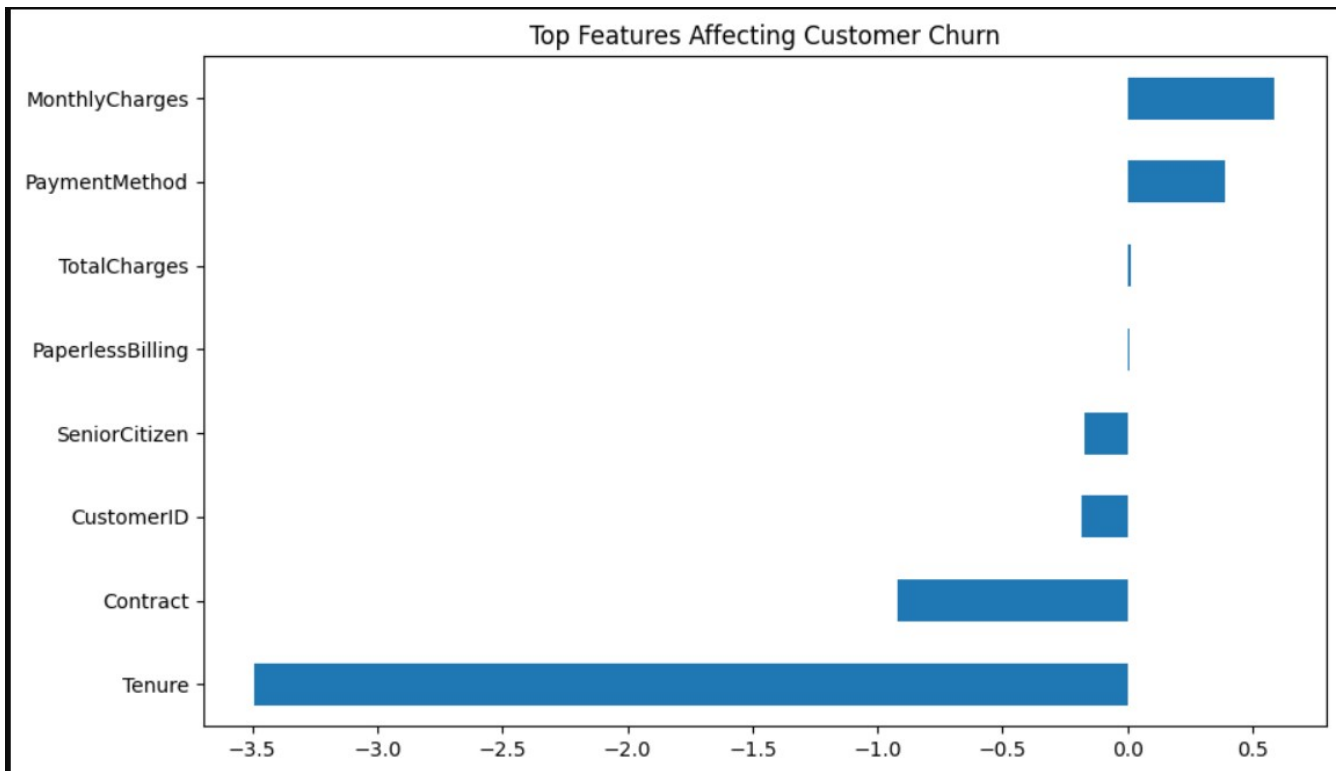
Confusion Matrix

```
[[132  2]
 [ 8  8]]
```



ROC Curve





True Negatives: 132 (Correctly identified non-churners)

- False Positives: 2 (Incorrectly identified non-churners as churners)
- False Negatives: 8 (Incorrectly identified churners as non-churners)
- True Positives: 8 (Correctly identified churners)

Top Features Affecting Churn :

The coefficients from the Logistic Regression model indicate the impact of each feature on customer churn:

- MonthlyCharges: 0.589
- PaymentMethod: 0.392
- TotalCharges: 0.014
- PaperlessBilling: 0.007
- SeniorCitizen: -0.174
- CustomerID: -0.184
- Contract: -0.921
- Tenure: -3.492